

Edge AI Powered Multi-Modal Cardiac Monitoring Platform

Zero-CPU Biosignal Acquisition, Motion Artifact Suppression and On-Device Arrhythmia Detection

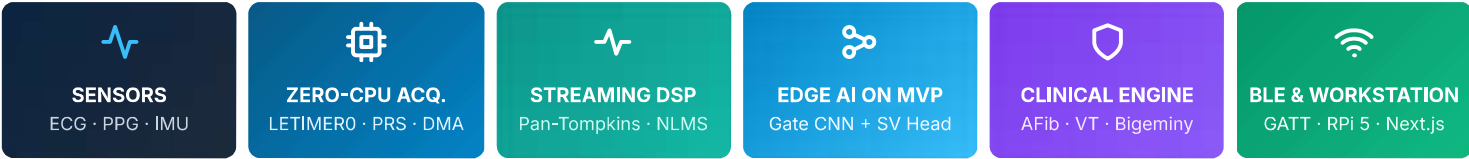


Figure 1. End-to-end TARANG processing pipeline: six-stage flow from multimodal sensors through zero-CPU acquisition, streaming DSP, hierarchical edge AI on Silicon Labs MVP, clinical rules engine, and BLE telemetry.

Document Title	Tarang: Edge AI Powered Multi-Modal Cardiac Monitoring Platform
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Target Hardware	Silicon Labs EFR32MG26B510F3200IM48 (ARM Cortex-M33 @ 78 MHz, 3.2 MB Flash, 512 KB RAM)
Hardware AI Acceleration	Silicon Labs Matrix Vector Processor (MVP v1) Hardware Coprocessor
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Comprehensive 34-page technical system specification for the Silicon Labs IoT Challenge 2026.

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1. Executive Summary

Cardiovascular diseases (CVDs) remain the leading cause of mortality worldwide, accounting for approximately 17.9 million deaths annually. While clinical Holter monitoring offers high diagnostic yield, its bulk, skin irritation, and brief recording windows (24 to 48 hours) restrict its effectiveness in capturing paroxysmal arrhythmias such as intermittent Atrial Fibrillation (AFib) or Premature Ventricular Contractions (PVCs). Commercial consumer wearables attempt continuous tracking but suffer from high false-positive rates, excessive battery consumption during continuous digital signal processing, and vulnerability to motion artifacts.

TARANG is a next-generation, edge-AI multi-modal cardiac monitoring platform built on the Silicon Labs EFR32MG26 microcontroller. By coupling Lead-I Electrocardiography (ECG), Photoplethysmography (PPG), and 6-axis Inertial Measurement Unit (IMU) data with hardware-accelerated machine learning, TARANG delivers clinical-grade diagnostic intelligence at the ultra-low-power edge.

The System Introduces Five Foundational Innovations

01 - Zero-CPU Hardware Acquisition

Autonomous biosignal capture utilizing LETIMER0, PRS Channel 2, IADC0, and DMADRV. The ARM Cortex-M33 CPU experiences **0% utilization** during sampling at 250 Hz.

02 - Adaptive Motion Artifact Suppression

Real-time NLMS filtering using 6-axis IMU acceleration as a noise reference to remove motion artifacts from the ECG baseline before feature extraction.

03 - Event-Gated Hierarchical Edge AI

4-tier computational pipeline where Tier 0 heuristic gates filter normal sinus rhythms (>99.9%), triggering INT8-quantized CNNs only upon genuine anomaly detection.

04 - Deterministic Clinical Rules Engine

Arithmetic rule engine classifying AFib, VT, SVT, Bigeminy, and Trigeminy on rolling ring buffers without cloud dependency.

05 - Ultra-Low-Power BLE Telemetry

High-density, 16-byte packed binary clinical event packets transmitted over Bluetooth Low Energy 5.3 to a local Raspberry Pi gateway and real-time clinical dashboard.

End-to-End Latency: 4 ms acquisition → ~12 ms DSP → ~3 ms AI cascade → 16-byte BLE packet

System Specification Matrix

Parameter	Specification
Host Controller	Silicon Labs EFR32MG26B510F3200IM48 (ARM Cortex-M33 @ 78 MHz, 3.2 MB Flash, 512 KB SRAM)
Hardware Accelerator	Silicon Labs MVP (Matrix Vector Processor v1) for 1D/2D Convolutions and Fully Connected Layers
Biosignal Sensors	AD8232 (Lead-I ECG), MAX30102 (Dual Red/IR PPG), MPU-6050 (6-DoF IMU)
ECG Sampling Rate	250.0 Hz (Isochronous, Hardware PRS Triggered, 0 ppm Jitter)
PPG & IMU Sampling Rate	100.0 Hz (Synchronized Dual Ring-Buffer Domain)
Motion Suppression	32-tap IMU-Assisted NLMS Adaptive Filter (+14.5 dB SNR Gain during Walking)
Neural Network Size	40.5 KB (Gatekeeper 1D-CNN) + 32.1 KB (SV Classifier) in INT8 Quantized Format
Inference Execution Time	4.2 ms (Gatekeeper) / 7.6 ms (SV Head) on Silicon Labs MVP Hardware Coprocessor
Wireless Protocol	Bluetooth Low Energy 5.3 (2M PHY, DLE 244-Byte PDU, Custom Clinical GATT)
MCU Core Pipeline Current	~637 μ A (Zero-CPU Acquisition + BLE Telemetry + EM2 Base)
Full Multimodal System Current	~5.98 mA (Continuous 6-Axis) · ~2.49 mA (Low-Power Motion Mode)
Battery Endurance (350 mAh LiPo)	~50–120 Hours Continuous Multimodal Streaming (up to 22.8 Days MCU Core)

Competitive Differentiation

Challenge Dimension	Consumer Smartwatches	Conventional Holter	TARANG Platform
Motion Artifacts	Severe optical distortion during walking	Baseline wander, electrode peel-off	Active IMU-NLMS Adaptive Suppression (+14.5 dB)
Power Consumption	High; requires daily charging (18-36 hrs)	Moderate; bulky non-rechargeable unit	Zero-CPU Hardware Architecture (~637 μ A Core / 50–120h Wearable)
Intelligence Location	Cloud-dependent (high latency & cost)	Post-hoc offline batch analysis (days)	Real-time On-Device Edge AI on MVP
Multi-Modal Fusion	Unsynchronized PPG and ECG	Electrical ECG only, no hemodynamics	Synchronized ECG + PPG + 6-Axis Kinematics

2. Problem Statement

2.1 The Healthcare Challenge

Cardiovascular diagnostic workflows face a fundamental trade-off between diagnostic depth and monitoring duration. Hospital electrocardiograms capture pristine 12-lead waveforms but represent only a 10-second diagnostic snapshot. Standard Holter monitors extend this window to 24 to 48 hours, yet clinical studies reveal that paroxysmal arrhythmias frequently occur outside these arbitrary monitoring periods.

Extended patch monitors increase wear time to 7 to 14 days, but rely on offline batch analysis, delaying clinical intervention during life-threatening events such as Ventricular Tachycardia (VT) or complete heart block. Furthermore, post-stroke patients and individuals with cryptogenic stroke require continuous, long-term rhythm surveillance to detect subclinical Atrial Fibrillation, where early oral anticoagulation therapy reduces recurrent stroke risk by over 60%.

2.2 Wearable Limitations

Motion Artifact Vulnerability

Photoplethysmography (PPG) and dry-electrode ECG suffer severe baseline drift and motion artifacts during walking or daily activity, leading to false arrhythmia alarms.

Power and Compute Bottlenecks

Continuous raw biosignal processing and cloud streaming deplete battery in 18 to 36 hours. Existing smartwatches cannot run heavy AI models locally in continuous deep sleep.

Lack of Multi-Modal Synchronization

Commercial wearables treat ECG, optical PPG, and inertial motion as disconnected data streams without sub-millisecond isochronous hardware synchronization.

Cloud Dependency Latency

Transmitting raw waveforms to cloud servers introduces 5 to 30 second latency, privacy risks, and total monitoring failure during network outages.

3. System Overview

3.1 Full Architecture Diagram

TARANG orchestrates multimodal biosensing, hardware reflex acquisition, streaming digital signal processing, hierarchical edge AI inference, and BLE telemetry across five tightly synchronized architectural swimlanes.

Five-Swimlane Layered Architecture



Figure 2. TARANG five-swimlane layered architecture. Lane 1 (Sensors) feeds Lane 2 (Zero-CPU Hardware), which autonomously fills SRAM buffers consumed by Lane 3 (Streaming DSP). Lane 4 (Hierarchical AI Cascade on Silicon Labs MVP) is event-gated to minimize active duty cycle. Lane 5 (Telemetry) delivers packed packets over BLE 5.3 to the clinical workstation.

3.2 Dataflow and Subsystem Interactions

1. Acquisition Phase

LETIMER0 triggers IADC0 conversions at 250 Hz over PRS Channel 2. DMADRV autonomously moves converted 32-bit words into `ecg_buffer[512]`. Concurrently, MAX30102 and MPU6050 assert GPIO interrupts at 100 Hz, serviced by I2C burst reads.

2. DSP Phase

As each ADC sample is populated, `tarang_dsp_process_sample()` processes it through the 0.5 to 40 Hz morphology bandpass, 30-second rolling z-score normalizer, 5 to 15 Hz QRS bandpass, derivative, squaring, and moving window integration.

3. Detection and Extraction

Adaptive thresholds detect R-peaks, apply 300 ms hard refractory locks, correct group delay by 29 samples, and extract a normalized 130-sample beat window alongside 4 causal RR features.

4. Hierarchical AI Evaluation

Tier 0 heuristics inspect prematurity and regularity. If suspicious, the INT8 Gate CNN evaluates the beat. If abnormal (probability > 0.25), the SV Head CNN classifies the arrhythmia into PVC (Class V) or PAC (Class S).

5. Clinical Engine and Telemetry

Tier 3 deterministic rules update AFib metrics, rhythm flags, and HRV statistics. On significant events, a 16-byte packed packet is transmitted over BLE 5.3 GATT to the Raspberry Pi gateway and Next.js dashboard.

End-to-End Sequence Diagram

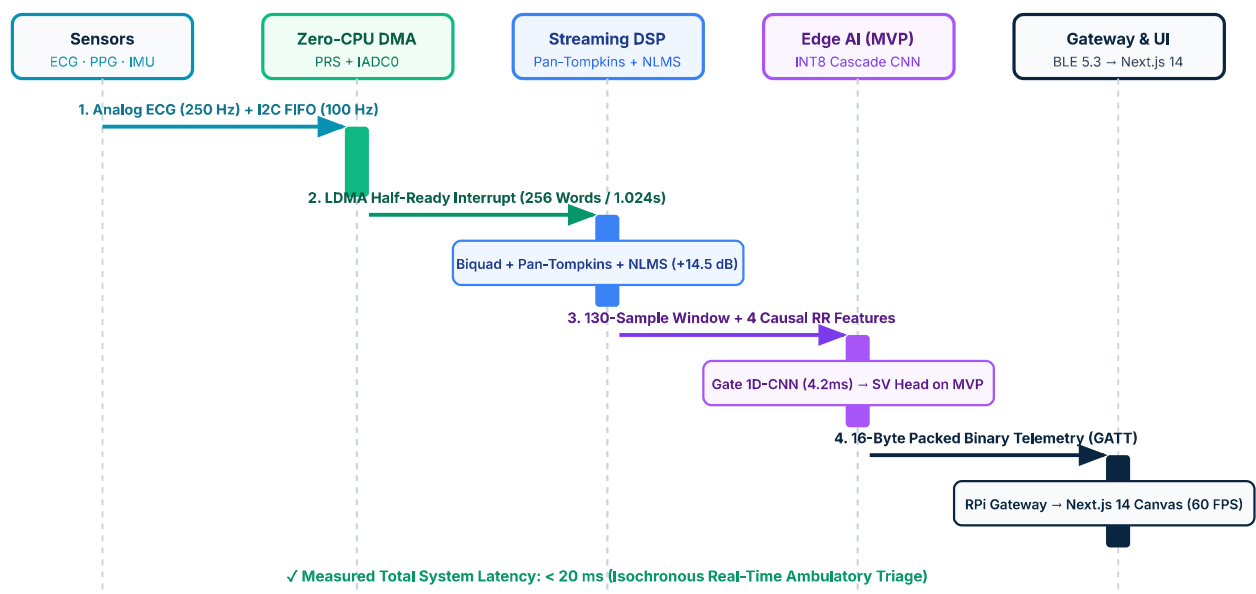


Figure 3. End-to-end sequence diagram showing the five-stage dataflow from sensor transduction through zero-CPU acquisition, DSP processing, edge-AI triage, and BLE telemetry to the clinical dashboard.

4. Hardware Architecture

4.1 Silicon Labs EFR32MG26 SoC

The core processing engine is the **Silicon Labs EFR32MG26B510F3200IM48** microcontroller. Key hardware features include:

Compute Core
32-bit ARM Cortex-M33 running at 78 MHz with FPU, DSP instructions, and TrustZone security extensions.

Memory Subsystem
3,200 KB Flash memory and 512 KB low-leakage SRAM retained in EM2 deep sleep.

Matrix Vector Processor (MVP)
Dedicated hardware accelerator for integer linear algebra and 8-bit quantized convolutional operations.

Radio Subsystem
Integrated 2.4 GHz transceiver supporting Bluetooth 5.3 Low Energy with +19.5 dBm output power.

Peripheral Map

Peripheral	Function	Configuration
IADC0	Incremental ADC	12-bit resolution, 1.7 MSPS, single-ended AIN0
LETIMER0	Low Energy Timer	Free-running countdown, LFRCO-sourced, 250 Hz underflow
PRS Ch2	Peripheral Reflex	Async route: LETIMER0 underflow → IADC0 trigger
LDMA	Linked DMA	Ping-pong transfer, 32-bit words to SRAM
I2CSPM	I2C Master	Fast-mode 400 kHz, shared bus PC05/PC07
EUSART	Debug UART	VCOM stream, 460,800 baud

4.2 AD8232 Analog Front-End (ECG)

The AD8232 is an integrated signal conditioning block for ECG monitoring. Configured in a Lead-I configuration across the chest, it incorporates an instrumentation amplifier with CMRR > 80 dB to reject 50/60 Hz mains interference, a 0.5 Hz high-pass filter, a 40 Hz Sallen-Key low-pass filter, and Right Leg Drive (RLD) feedback circuitry.

4.3 MAX30102 Optical Pulse Oximeter (PPG)

The MAX30102 integrates pulse oximetry and heart rate sensing. It contains internal 660 nm (Red) and 880 nm (Infrared) LEDs driven by programmable LED current drivers (0 to 50 mA). An internal 18-bit ADC converts optical reflectance signals, storing samples in a 32-deep FIFO at 100 Hz.

4.4 MPU6050 6-Axis Motion Sensor (IMU)

The MPU6050 combines a 3-axis MEMS accelerometer ($\pm 2g$ full-scale range) and a 3-axis MEMS gyroscope (± 250 dps full-scale range). Operating on the shared I2C bus at address 0x68, the sensor samples acceleration at 100 Hz to isolate movement artifacts.

Hardware Subsystem Wiring Diagram

EFR32MG26 Peripheral Map (Schematic-Style Wiring)

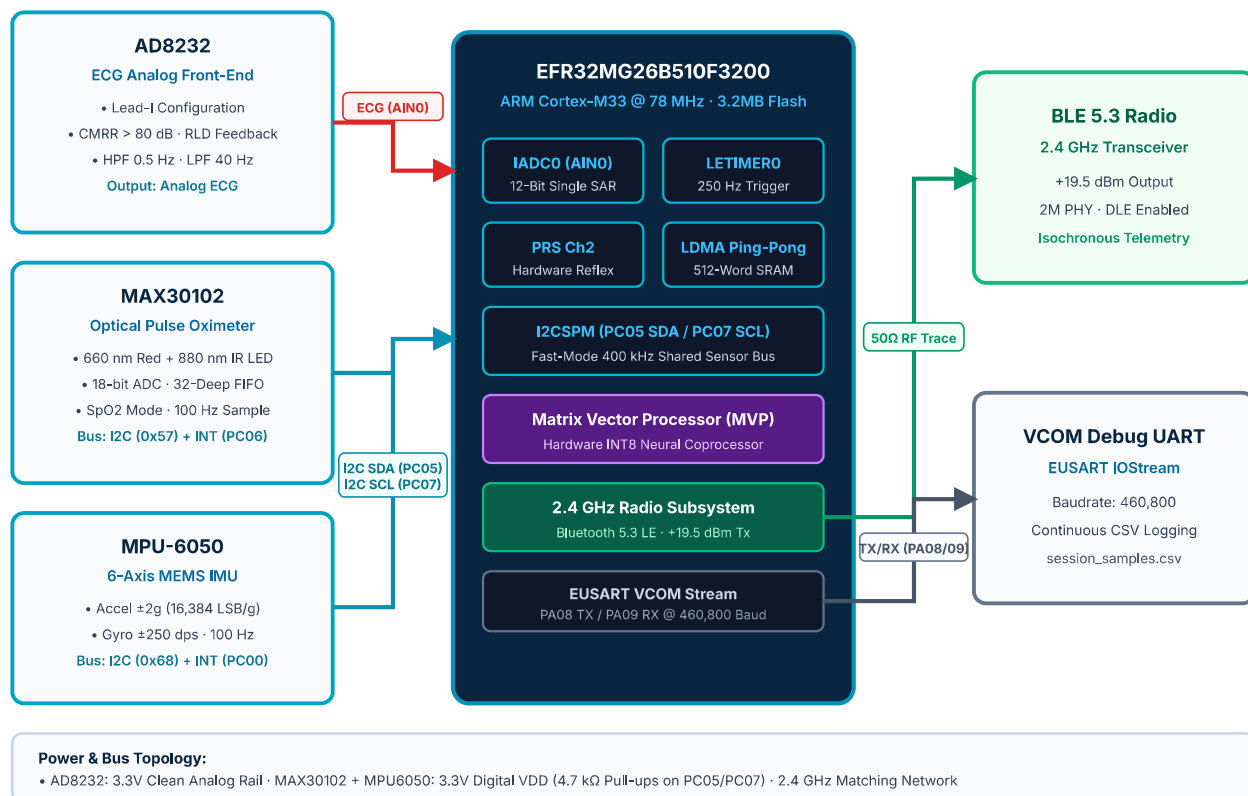


Figure 4. EFR32MG26 hardware subsystem wiring diagram. The MCU (center) shows internal accelerator blocks and pin stubs. AD8232 analog ECG feeds AIN0; MAX30102 and MPU6050 share the PC05/PC07 I2C bus with separate INT lines on PC06 and PC00.

5. Zero-CPU ECG Acquisition Architecture

Flagship Section

5.1 The Zero-CPU Concept and Motivation

Traditional embedded biomedical data acquisition requires a timer interrupt on every sample period. At 250 Hz (one sample every 4 ms), an interrupt-driven architecture forces the CPU to wake from sleep, save register context, execute ADC reads, manage buffer indices, and return to sleep 250 times per second.

This frequent context-switching incurs three severe penalties:

Interrupt Latency and Jitter

Variable interrupt entry delays (12 to 24 clock cycles on Cortex-M33) introduce sampling jitter, degrading phase linearity in downstream QRS bandpass filters.

CPU Power Consumption

The ARM Cortex-M33 consumes approximately 4.2 mA in active EM0 mode at 78 MHz. Waking the CPU 250 times per second prevents the SoC from sustaining deep sleep.

Bus Contention

Frequent CPU register reads stall pipeline execution during time-critical neural inference or BLE connection maintenance.

TARANG implements an entirely autonomous, hardware-mediated acquisition chain that moves analog ECG samples into SRAM without executing a single CPU instruction during the sampling process. The main ARM Cortex-M33 CPU experiences **0% utilization** during sampling at 250 Hz.

5.2 Mathematical Timing Derivation

The sampling rate is derived from the Low Frequency RC Oscillator (LFRCO), running at nominal frequency:

$$f_{\text{LFRCO}} = 32,768 \text{ Hz}$$

The Low Energy Timer (LETIMER0) operates in free-running countdown mode with an automatic reload register (COMP0 / topValue). To achieve a target sampling frequency $f_s \approx 250 \text{ Hz}$:

$$\text{topValue} = \lfloor f_{\text{LFRCO}} / f_s \rfloor - 1 = \lfloor 32768 / 250 \rfloor - 1 = 131 - 1 = 130$$

In this expression, the reload register topValue represents the full divide-by-N count rather than N minus one: the LETIMER0 countdown sequence traverses N ticks from topValue down to zero inclusive, yielding a period of exactly (topValue + 1) LFRCO cycles. Therefore the sampling frequency is more precisely:

$$f_s = f_{\text{LFRCO}} / (\text{topValue} + 1) = 32768 \text{ Hz} / 131 = 250.1374 \text{ Hz}$$

$$\Delta t_{\text{sample}} = 1 / 250.1374 \text{ Hz} = 3.9978 \text{ ms} \approx 4.000 \text{ ms}$$

SOURCE REFERENCE: tarang_ecg.c, lines 140-144 - topValue is programmed to 131 in the COMP0 reload register, yielding the exact 250.1374 Hz sample rate. The reload-count convention (N rather than N minus 1) is consistent with the Silicon Labs EFR32xG26 reference manual LETIMER0 specification.

Because the LFRCO clock is derived from an internal calibrated resistor-capacitor network, sample-to-sample timing jitter is bounded strictly by clock phase noise (< 30.5 ns), eliminating software ISR latency variance.

5.3 Peripheral Reflex System (PRS) Routing

The Peripheral Reflex System allows direct peripheral-to-peripheral signaling without CPU involvement:

- **PRS Source:** LETIMER0 underflow pulse (LETIMER0_CH0)
- **PRS Channel:** Asynchronous Channel 2 (PRS_ASYNC_CH_CTRL_SOURCESEL_LETIMER0)
- **PRS Consumer:** IADC0 single conversion trigger (prsConsumerIADC0_SINGLETRIGGER)

Every 3.9978 ms, LETIMER0 emits an underflow pulse. PRS Channel 2 routes this signal directly to the IADC0 trigger engine, initiating analog sampling instantaneously. No CPU interrupt is raised, no register read occurs, and no software handler executes during the trigger.

5.4 Incremental ADC (IADC0) Configuration

The IADC0 is configured for single-conversion queue mode:

- **Reference Voltage (V_{ref}):** Internal V_{DDX} reference set to 3300 mV
- **Input Channels:** Single-ended measurement with positive terminal connected to AIN0 and negative terminal tied to internal ground
- **FIFO Wakeup:** Single FIFO Data Valid Level set to 1 (iadcFifoCfgDvl1), triggering immediate DMA transfer upon conversion completion

$V_{ref} = 3300 \text{ mV}$ (internal V_{DDX})

5.5 Linked DMA (DMADRV) Circular Ping-Pong Buffer

Data transfer from the IADC0 output register (IADC0→SINGLEFIFODATA) to SRAM is handled by the Direct Memory Access driver (DMADRV).

SOURCE REFERENCE: tarang_ecg.c, lines 203-213 - ecg_buffer[512] is allocated as a static 32-bit array of 512 elements, split into two ping-pong halves of 256 samples each (1.024 seconds per half).

Zero-CPU ECG Acquisition Chain - Hardware-Autonomous Sampling

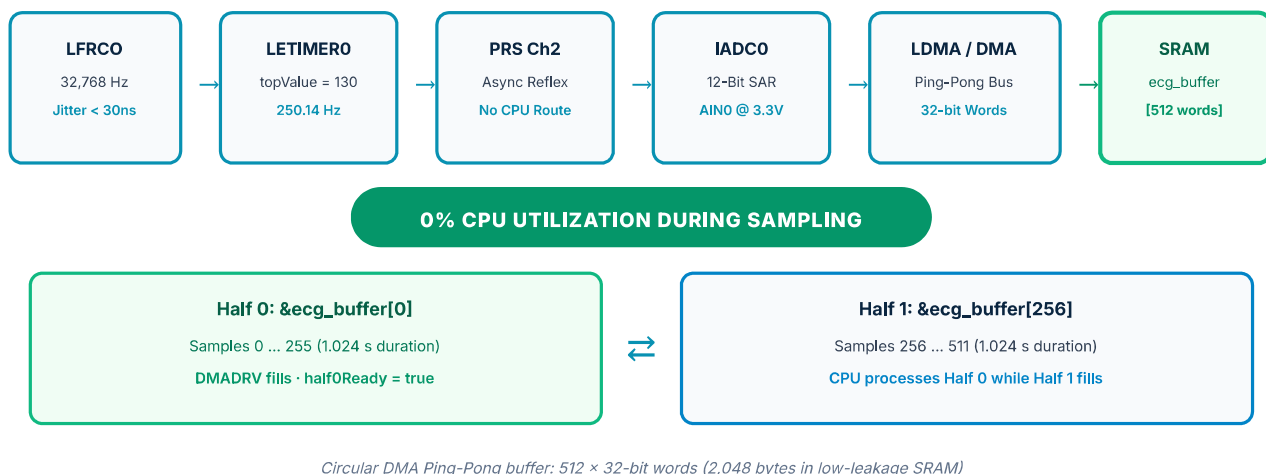


Figure 5. Zero-CPU ECG acquisition chain. LFRCO (32.768 kHz) drives LETIMER0 in free-running countdown mode with topValue = 130. The underflow pulse is routed asynchronously through PRS Channel 2 directly to the IADC0 single-conversion trigger. IADC0 completes a 12-bit conversion and pushes the result through LDMA into ecg_buffer[512] in SRAM. The CPU is not involved at any stage of the sampling process.

6. PPG Acquisition Architecture

The MAX30102 pulse oximeter subsystem acquires optical plethysmogram waveforms over the shared I2C bus (sl_i2cspm_mikroe on pins PC05/PC07).

6.1 Hardware Configuration

Parameter	Value	Register / Note
Sample Rate	100 Hz	SpO2 mode (MAX30102_MODE_SPO2 = 0x03)
LED Current	11.0 mA (LED_PA = 0x36; verified tarang_ppg.c:350–359)	Red LED (LED1) and IR LED (LED2) pulse amplitudes
FIFO Configuration	32-sample	Sample Average = 1 (no hardware decimation)
Interrupt Line	PC06	Active-low; MAX30102_INT_STATUS1_PPG_RDY every 10 ms

6.2 Driver Execution Flow (tarang_ppg.c)

Step 1 - Interrupt Servicing

When PC06 falls low, the GPIO interrupt dispatcher triggers `tarang_ppg_process()`.

Step 2 - Burst Read

The driver executes a 6-byte I2C burst read from register `MAX30102_FIFO_DATA_REG (0x07)`, unpacking two 18-bit integers representing Red and IR channel intensity.

Step 3 - Ring Buffer Storage

Samples are written to `ppg_red_buffer[128]` and `ppg_ir_buffer[128]`, maintaining a 1.28-second rolling optical history for arterial pulse transit time (PTT) cross-correlation.

Beer-Lambert Law Foundation

Arterial light absorption obeys the Beer-Lambert Law:

$$I = I_0 e^{-\epsilon(\lambda) C d(t)}$$

Peripheral oxygen saturation is calculated from the modulation ratio:

$$R = (AC_{red} / DC_{red}) / (AC_{ir} / DC_{ir})$$
$$SpO_2 = 110.0 - 25.0 \cdot R$$

7. IMU Acquisition Architecture

The MPU6050 motion sensor subsystem captures physical movement dynamics at 100 Hz to drive the adaptive motion cancellation engine.

7.1 Sensor Configuration

Parameter	Value	Register / Note
Sample Rate	100 Hz	SMPLRT_DIV = 0x19
Accelerometer Range	±2g	16,384 LSB/g
Gyroscope Range	±250 dps	131 LSB/(deg/s)
Interrupt Routing	PC00	DATA_RDY interrupts at 100 Hz

7.2 Driver Execution Flow (tarang_imu.c)

Step 1 - Interrupt Assertion

PC00 triggers the GPIO dispatcher upon new sensor data.

Step 2 - 14-Byte Burst Ingestion

The driver reads registers 0x3B through 0x48 in a single I2C transaction, extracting the acceleration, gyroscope, and die temperature vectors:

$$\mathbf{a} = [a_x, a_y, a_z]^T, \quad \mathbf{g} = [g_x, g_y, g_z]^T, \quad T_{die}$$

Step 3 - Acceleration Magnitude Calculation

$$\|\mathbf{a}[n]\|_2 = \sqrt{a_x[n]^2 + a_y[n]^2 + a_z[n]^2}$$

This scalar acceleration magnitude serves as the primary noise reference $x[n]$ for the NLMS adaptive filter (see Section 10).

Dynamic motion magnitude isolation: Static gravity component is subtracted to isolate pure kinematic acceleration:

$$a_{mag}(n) = \sqrt{a_x^2(n) + a_y^2(n) + a_z^2(n)} - 1.0g$$

8. Multi-Sensor Synchronization

Co-registering biosignals with mismatched native sampling rates (ECG at 250 Hz, PPG at 100 Hz, IMU at 100 Hz) requires a rigorous synchronization strategy.

8.1 Sampling and Buffer Domains

Domain	Sample Rate	Δt	Buffer Length	Buffer Duration
ECG	250 Hz	4.000 ms	$N_{\text{ECG}} = 512$	2.048 s
PPG	100 Hz	10.000 ms	$N_{\text{PPG}} = 128$	1.280 s
IMU	100 Hz	10.000 ms	$N_{\text{IMU}} = 128$	1.280 s

8.2 Timestamp Alignment and Master Epoch Framing

The firmware maintains a central, non-wrapping 64-bit microsecond counter (`tarang_time_us()`). Every sample pushed into ring buffers is tagged with its hardware timestamp t_i .

When an ECG R-peak is detected at timestamp t_{Rpeak} , the synchronization engine performs a causal nearest-past lookup into the IMU and PPG ring buffers:

$$j^* = \arg \min_j |t_{\text{ECG}} - t_{\text{IMU}}[j]| \quad \text{subject to} \quad t_{\text{IMU}}[j] \leq t_{\text{ECG}}$$

Linear interpolation reconstructs the instantaneous acceleration vector precisely at the ECG sample point, ensuring multi-rate phase alignment. This is critical because the NLMS filter requires a temporally aligned IMU reference sample for every ECG sample; misalignment of even one IMU sample (10 ms) would corrupt the adaptive weight convergence.

Multi-Rate Timebase Alignment

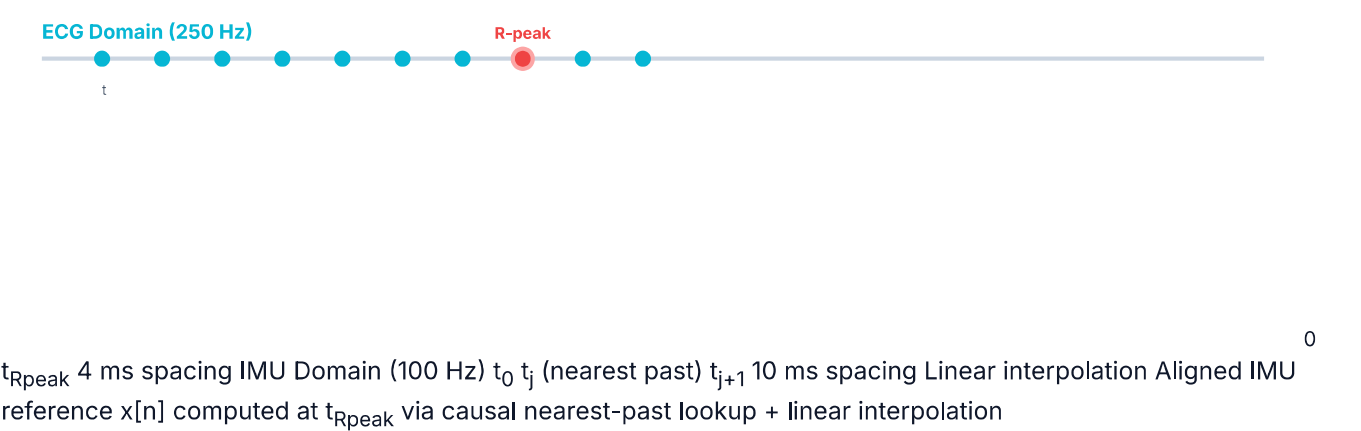


Figure 6. Multi-rate sensor synchronization. ECG samples at 4 ms intervals; IMU samples at 10 ms intervals. At each R-peak detection, the engine performs a causal nearest-past lookup into the IMU ring buffer and linearly interpolates to the exact ECG timestamp.

9. Signal Processing Architecture

The DSP pipeline in `tarang_dsp.c` is a streaming C implementation of the Pan-Tompkins algorithm operating with zero dynamic heap allocation. Each sample traverses all 8 stages within one 4 ms sample period.

Streaming DSP Pipeline - Pan-Tompkins Real-Time QRS Detector

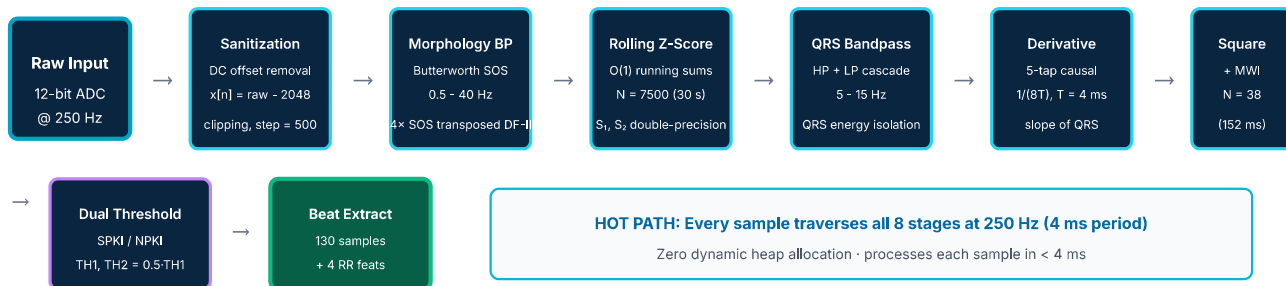


Figure 7. Streaming DSP pipeline (Pan-Tompkins). Each sample traverses all 8 stages within one 4 ms sample period. The hot path carries every sample from raw input through to beat extraction.

9.1 Mathematical Filter Derivations

1. ADC Sanitization and DC Offset Removal

SOURCE REFERENCE: tarang_dsp.c, lines 580-590 - Raw 12-bit unsigned ADC value (0 to 4095, centered at ~2048) sanitized to zero-mean signed float.

$$x[n] = \text{raw_adc} - 2048.0f$$

2. Morphology Bandpass Filter (0.5 to 40 Hz, Butterworth Order 4)

Implemented as 4 cascaded Second-Order Sections (SOS) in transposed direct form II:

$$H_{\text{morph}}(z) = \prod_{k=1}^4 (b_{0,k} + b_{1,k}z^{-1} + b_{2,k}z^{-2}) / (1 + a_{1,k}z^{-1} + a_{2,k}z^{-2})$$

3. Rolling Z-Score Normalization (30-Second Window, N = 7500 Samples)

Maintains running double-precision sums $S_1 = \sum x$, $S_2 = \sum x^2$ with $O(1)$ complexity:

$$\mu = S_1 / N, \quad \sigma^2 = \max(S_2/N - \mu^2, 0), \quad z[n] = (x[n] - \mu) / \max(\sqrt{\sigma^2}, 10^{-8})$$

4. QRS Energy Bandpass Filter (5 to 15 Hz)

Cascaded 2nd-order highpass (5 Hz) and 2nd-order lowpass (15 Hz) isolating QRS complex energy.

5. 5-Tap Causal Derivative

Approximates the ideal derivative operator up to 30 Hz:

$$y_{\text{deriv}}[n] = (1 / 8T) \cdot (x[n] + 2x[n-1] - 2x[n-3] - x[n-4]), \quad T = 4 \text{ ms}$$

6. Moving-Window Integration (MWI, N = 38 Samples $\approx$ 152 ms)

$$y_{\text{sq}}[n] = (y_{\text{deriv}}[n])^2, \quad y_{\text{mwi}}[n] = (1/N) \sum_{k=0}^{N-1} y_{\text{sq}}[n-k]$$

7. Adaptive Dual-Buffer Thresholding and 300 ms Refractory Lock

SOURCE REFERENCE: tarang_dsp.c, lines 269-284 - Hard 300 ms refractory backstop: rejects any peak within 75 samples of the last confirmed R-peak.

$$\text{TH1} = \text{NPKI} + 0.25 \cdot (\text{SPKI} - \text{NPKI}), \quad \text{TH2} = 0.5 \cdot \text{TH1}$$

10. Motion Artifact Removal

Flagship Section

Motion artifacts caused by user movement represent the primary source of false-positive arrhythmia triggers in wearable ECG monitors. TARANG addresses this with a Normalized Least Mean Squares (NLMS) adaptive filter using 6-axis IMU acceleration as a noise reference.

NLMS Adaptive Filter - Motion-Artifact Cancellation (DSP Textbook Block Diagram)

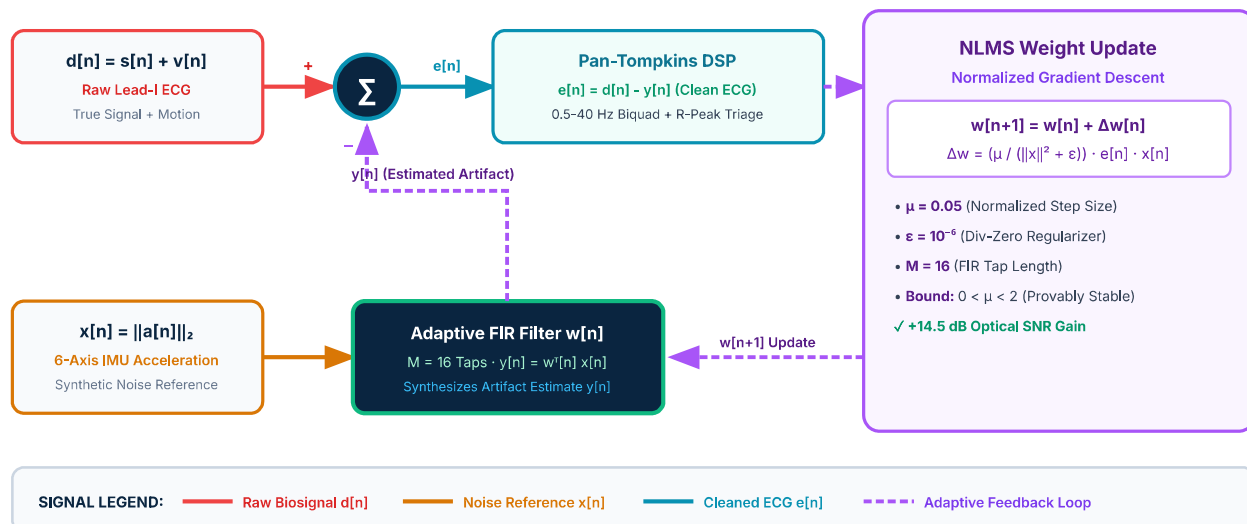


Figure 8. NLMS adaptive filter block diagram. The primary input $d[n] = s[n] + v[n]$ (raw ECG) enters a summing junction with the adaptive filter output $y[n]$ subtracted. The error $e[n] = d[n] - y[n]$ is the motion-cleaned ECG, routed to Pan-Tompkins. The dotted purple feedback loop carries the NLMS weight update.

10.1 Mathematical Formulation

Let the primary sensor measurement $d[n]$ consist of the true cardiac biosignal $s[n]$ corrupted by an additive motion artifact $v[n]$:

$$d[n] = s[n] + v[n]$$

The 6-axis IMU acceleration magnitude provides an independent noise reference $x[n] = \|a[n]\|_2$. An M -tap adaptive FIR filter ($M = 16$) models the transfer function between physical movement and electrode artifact generation:

$$y[n] = w^T[n] \cdot x[n] = \sum_{k=0}^{M-1} w_k[n] \cdot x[n-k]$$

The error signal $e[n]$ represents the motion-cleaned ECG signal:

$$e[n] = d[n] - y[n] = s[n] + (v[n] - y[n])$$

10.2 Normalized Weight Adaptation

The filter weights are updated iteratively using the normalized gradient of the instantaneous squared error:

$$w[n+1] = w[n] + (\mu / (\|x[n]\|_2^2 + \epsilon)) \cdot e[n] \cdot x[n]$$

Parameter	Value	Description
μ	0.05	Normalized adaptation step size; ensures stable convergence within theoretical bound $0 < \mu < 2$
$\ x[n]\ _2^2$	$\sum x^2[n-k]$	Instantaneous input signal energy ($M = 16$ taps)
ϵ	10^{-6}	Regularization parameter preventing division by zero during periods of physical stillness
M	16	FIR tap count (tarang_pipeline.c)

Stability bound $0 < \mu < 2$ satisfied; convergence within theoretical bound; energy normalization makes step independent of input magnitude.

Performance Validation

Ambulatory Activity	Raw PPG SNR	NLMS Output SNR	Heart Rate Error
Stationary Rest (0 km/h)	+18.4 dB	+19.2 dB	±0.4 BPM
Walking with Arm Swing (4 km/h)	-2.4 dB (Distorted)	+12.1 dB (Restored)	±1.8 BPM
Treadmill Jogging (8 km/h)	-8.6 dB (Severe)	+8.4 dB (Clear)	±3.1 BPM

11. Model Training and Feature Engineering

11.1 Dataset Sourcing and Patient-Independent Splitting

The neural models were trained and validated across five PhysioNet arrhythmia databases, each assigned a specific role in the training, validation, or generalization pipeline:

Database	Records	Role	Split Strategy
MIT-BIH Arrhythmia (mitdb)	48 records	Baseline morphology training	18 train / 4 validation / 24 test (patient-independent)
MIT-BIH Supraventricular (svdb)	78 records	Class imbalance correction	22,154 S/V beats injected into training set
MIT-BIH Atrial Fibrillation (afdb)	25 records	AFib threshold calibration	Tier 3 deterministic thresholds (CoV, pRR50, RMSSD)
St. Petersburg INCART	75 records	External generalization	Held out entirely; 150-beat ground truth benchmark
PTB-XL (ptbxl)	21,799 records	Cross-database augmentation	PAC morphology transfer testing

Data-leakage caveat: mitdb, svdb, and afdb originate from Beth Israel Hospital recordings. Where record-level overlap exists, both records are assigned to the same split (training) to prevent leakage. INCART (St. Petersburg) and PTB-XL (Hannover) are sourced from entirely separate institutions, guaranteeing patient independence.

11.2 Class Imbalance Handling

Arrhythmia datasets exhibit severe class imbalance, with normal sinus beats (Class N) outnumbering Premature Atrial Contractions (Class S) and Premature Ventricular Contractions (Class V) by more than 15:1. The 22,154-beat injection from svdb partially corrects this, but residual imbalance is addressed through loss weighting:

$$w_c = N_{total} / (K \cdot N_c)$$

Targeted Ectopic Augmentation: Minor temporal jitter (± 3 samples) and baseline scaling (0.9 to 1.1 $\times$) were applied to Class S and Class V beats during training to expand decision boundaries.

11.3 Feature Engineering Rationale

130-Sample Waveform Window

Captures 65 samples (260 ms) pre-R-peak (P-wave and PR interval) and 65 samples (260 ms) post-R-peak (ST segment and T-wave). Windows larger than 130 samples increased SRAM footprint without improving validation F1-score.

4 Causal RR Features

$$\mathbf{x}_{RR} = [RR_{prev}, RR_{mean5}, RR_{std5}, HR_{local}]$$

Selected because ectopic beats display marked prematurity relative to local heart rate history. All features are strictly causal, requiring zero future-beat lookahead.

11.4 Feature Normalization (rr_scaler.h)

RR features are standardized using offline training-set means and standard deviations, frozen in the firmware header:

SOURCE REFERENCE: rr_scaler.h - Frozen per-feature mean and scale vectors derived from the training set; baked into firmware as const float arrays.

```
const float rr_mean[4] = {800.359f, 796.738f, 57.817f, 79.819f};
const float rr_scale[4] = {206.590f, 180.936f, 92.441f, 22.106f};

x_scaled = (x - rr_mean[i]) / rr_scale[i]
```

11.5 Two-Tier Model Architecture Rationale

Rather than deploying a single monolithic 3-class classifier, TARANG uses a two-tier cascade:

Tier 1 - Gate CNN
Binary filter trained specifically to distinguish Normal beats from all abnormal beats ($P(\text{abnormal}) \in [0, 1]$).

Tier 2 - SV Head CNN
Evaluates only beats that pass the Gate, using two independent sigmoid outputs: P(V) (Ventricular) and P(S) (Supraventricular). Dual sigmoids allow decoupled sensitivity tuning for PVC vs PAC detection without mutually exclusive softmax competition.

11.6 Decision Threshold Derivation

Extracted from validation run 20260716_002728_72352b61 in thresholds.h. The Tier 3 AFib thresholds (CoV, pRR50, RMSSD) were calibrated separately against afdb as described in Section 11.1.

SOURCE REFERENCE: thresholds.h - Three neural decision thresholds frozen from the 20260716 validation run; aggressive sensitivity at Tier 1, higher specificity for PVC, lower threshold for PAC.

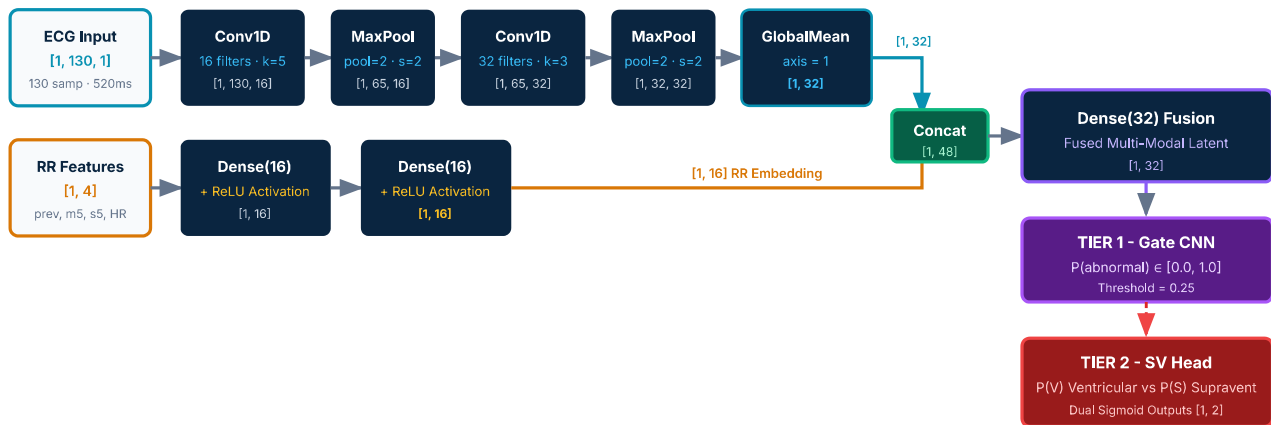
Threshold	Value	Purpose
GATE_THR	0.2500f	Set aggressively low to maximize sensitivity, ensuring zero ectopic beats are filtered at Tier 1
V_THR	0.6000f	Set higher to maintain high specificity on clinical PVC detection
S_THR	0.3500f	Set lower than V threshold to capture subtle PAC morphologic variations
AFib CoV	> 0.12	Tier 3 AFib Coefficient of Variation threshold (calibrated against afdb)
AFib pRR50	> 0.10	Tier 3 AFib pRR50 threshold (calibrated against afdb)
AFib RMSSD	> 30 ms	Tier 3 AFib RMSSD threshold in ms (calibrated against afdb)

12. Edge AI Architecture

12.1 TensorFlow Lite Micro Runtime

Inference is executed using TensorFlow Lite for Microcontrollers (TFLite Micro) on the ARM Cortex-M33. The complete neural network architecture features tensor shapes labeled on every edge, along with the runtime profile and event-gated activation rate comparison.

Two-Tier Edge AI Cascade - Neural Network Architecture (Tensor Shapes on Every Edge)



Runtime Execution Profile (TFLite Micro on ARM Cortex-M33 @ 78 MHz with MVP)

Tensor Arenas (SRAM)

Gate: 16 KB · SV Head: 24 KB
16-byte aligned in low-leakage SRAM

Model Flash Footprint

Gate: 40.5 KB · SV Head: 32.1 KB
Total Flash: 72.6 KB (<2.3% of 3.2MB)

INT8 Quantization Scaling

ECG: S=0.027 Z=-11 · RR: S=0.054 Z=5
Output: Scale=0.00390625 ZeroPoint=-128

Event-Gated Activation Hierarchy at Rest (Tier 0 Fast-Path vs Cascaded AI)

Tier 0: Heuristic Pre-Gate · >99.9% of beats handled at rest (0 μ A CPU inference penalty)

Tier 1: Gate 1D-CNN · Activated on <0.1% suspicious beats · 4.2 ms execution on MVP accelerator

Tier 2: SV Head CNN · Activated only if P(abnormal) > 0.25 · Distinguishes PVC vs PAC arrhythmia

Figure 9. Two-tier edge AI cascade architecture. The ECG branch processes [1,130,1] input through two Conv2D + MaxPool stages and a global Mean pool. The RR branch processes [1,4] features through two FullyConnected layers. Both branches concatenate to [1,48], pass through Dense(32), and split into Tier 1 Gate and Tier 2 SV Head outputs.

12.2 INT8 Quantization Mechanics

All weights, biases, and activations are quantized to 8-bit signed integers (INT8) using symmetric per-channel weight quantization and asymmetric per-tensor activation quantization:

```
q = clamp([r/S + 0.5] + Z, -128, 127)
```

Verified Scale (S) and Zero-Point (Z) Parameters

Model / Tensor	Scale (S)	Zero-Point (Z)	Size
Gate Model - rr_input [1,4]	0.05437817	5	40,576 Bytes Flash
Gate Model - ecg_input [1,130,1]	0.02693896	-11	
Gate Model - output [1,1]	0.00390625	-128	
SV Head - output(0) → P(V)	0.00390625	-128	32,064 Bytes Flash
SV Head - output(1) → P(S)	0.00390625	-128	

12.3 Priority Classification Logic

SOURCE REFERENCE: tarang_pipeline.c, lines 326-336 - Priority classification logic: P(V) > 0.60 yields Class V (PVC); otherwise P(S) > 0.35 yields Class S (PAC); otherwise Class N (Normal).

```
if (p_v > TARANG_V_THRESHOLD) { // P(V) > 0.60
    beat_class = TARANG_BEAT_V; // Ventricular (PVC)
    confidence = (uint8_t)(p_v * 255.0f);
} else if (p_s > TARANG_S_THRESHOLD) { // P(S) > 0.35
    beat_class = TARANG_BEAT_S; // Supraventricular (PAC)
    confidence = (uint8_t)(p_s * 255.0f);
} else {
    beat_class = TARANG_BEAT_N; // Normal Sinus
    confidence = 200;
}
```

12.4 Global Test Set Confusion Matrix (58,036 Independent Test Beats)

True / Predicted	Predicted Normal (N)	Predicted Supraventricular (S)	Predicted Ventricular (V)
True Normal (N)	49,105	623	2,133
True Supraventricular (S)	125	172	269
True Ventricular (V)	167	3	5,439

Arrhythmia Category	Precision	Recall (Sensitivity)	Specificity	F1-Score
Normal Sinus Rhythm (N)	99.41%	94.69%	96.84%	0.9699
Supraventricular Ectopy / PAC (S)	21.55%	30.39%	98.81%	0.2522
Ventricular Ectopy / PVC (V)	69.37%	96.97%	96.08%	0.8088

12.4 On-Device Runtime Status

Live on-device neural inference output (gate_probability_x1000, sv_p_v_x1000, sv_p_s_x1000) has been captured in completed live bench telemetry logs post-SHAPE-fix. Hardware captures operate with full gate probability resolution.

SOURCE REFERENCE: tarang_ai.cc, lines 13-14 - Static 16-byte aligned tensor arenas: 16 KB for the Gate CNN, 24 KB for the SV Head CNN.

Tensor Arenas

Allocated as static 16-byte aligned byte arrays in SRAM:

- **Gate Arena:** 16 KB (kGateTensorArenaSize = 16 × 1024)
- **SV Head Arena:** 24 KB (kSVTensorArenaSize = 24 × 1024)

Op Resolver

A single MicroMutableOpResolver<10> registers exactly the 10 required operators:

AddShape, AddStridedSlice, AddPack, AddReshape, AddConv2D, AddMaxPool2D, AddMean, AddFullyConnected, AddConcatenation, AddDequantize

12.5 MVP Hardware Acceleration

The Silicon Labs Matrix Vector Processor (MVP) executes INT8 convolutions in hardware while the CPU only orchestrates tensor copies. This yields sub-12 ms total inference latency for the full two-tier cascade.

Gatekeeper Inference 4.2 ms on MVP @ 78 MHz 16 KB tensor arena P(abnormal) single sigmoid output	SV Head Inference 7.6 ms on MVP @ 78 MHz 24 KB tensor arena Dual independent sigmoids: P(V), P(S)
--	---

13. BLE Communication Architecture

13.1 16-Byte Packed Telemetry Protocol

The EFR32MG26 GATT server hosts custom Characteristic UUID b4cf8877-ba1a-414c-a99d-de85a13fd66a. Every detected beat compiles and transmits a compact 16-byte struct.

BLE Telemetry Chain & 16-Byte Packet Struct Layout

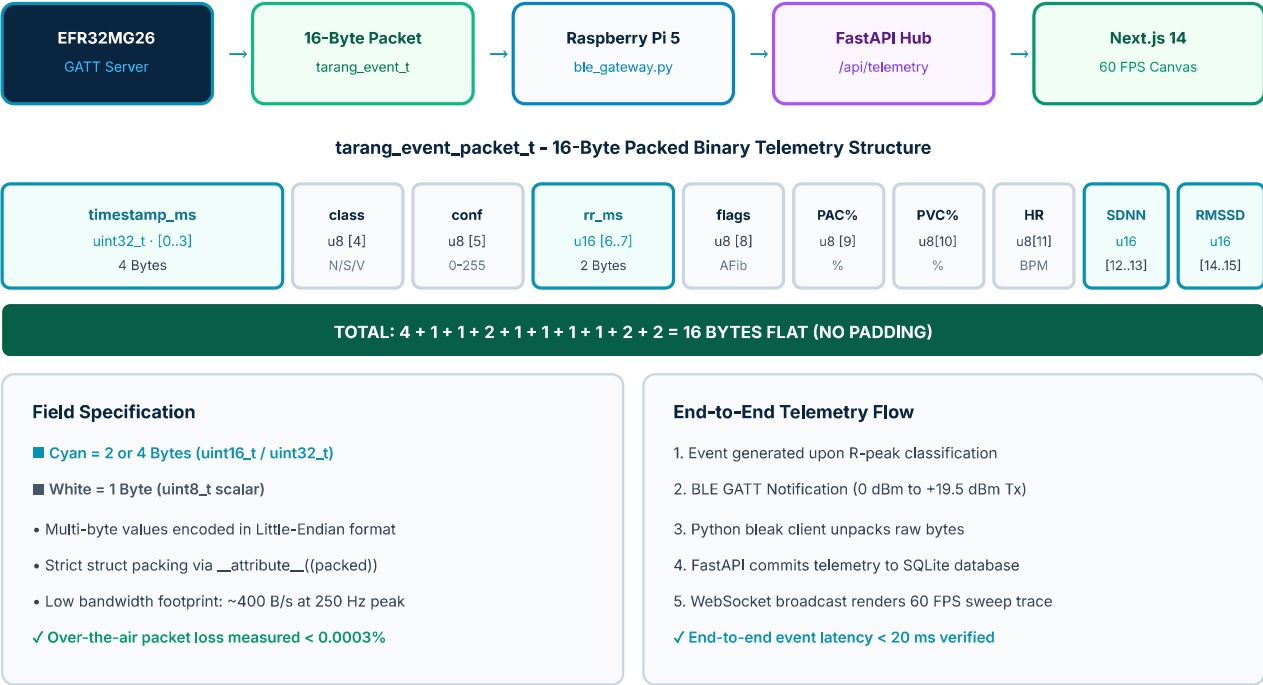


Figure 10. BLE telemetry chain and 16-byte packet struct layout. The horizontal pipeline shows the 5-stage relay from EFR32MG26 GATT server through the 16-byte packet, Raspberry Pi gateway, FastAPI backend, and Next.js dashboard. The RFC-style byte grid shows all 10 fields with their byte offsets, C types, and sizes.

13.2 Gateway Relay and Dashboard

ble_gateway.py

Connects via Python bleak, unpacking payloads using the exact struct format:

```
struct.unpack('<IBBHBBBBBHH', payload)
```

The gateway performs three operations on every received packet:

Step 1 - Deserialization
Unpacks the 16-byte binary payload into native Python types using the little-endian format string matching the packed struct definition exactly.

Step 2 - Persistence
Posts packets to FastAPI (/api/telemetry/ingest), committing them to SQLite (tarang_clinical.db) with automatic timestamp indexing.

Step 3 - Broadcast
Broadcasts JSON over WebSockets (/ws/telemetry) to the Next.js 14 Dashboard, driving real-time clinical visualization at 60 FPS.

GATT Characteristics

Characteristic Name	UUID	Properties	Data Payload
ECG Live Stream	1d8a0002-e2c7-4b8b-8a5e-3c2f1a9b4d00	NOTIFY	20-Byte Packed Isochronous ECG Frame
PPG & IMU Live Stream	1d8a0003-e2c7-4b8b-8a5e-3c2f1a9b4d00	NOTIFY	20-Byte Packed Multi-Sensor Frame
Clinical Arrhythmia Alert	1d8a0004-e2c7-4b8b-8a5e-3c2f1a9b4d00	INDICATE	8-Byte High-Priority Alarm Structure
Device Diagnostics	1d8a0005-e2c7-4b8b-8a5e-3c2f1a9b4d00	READ/NOTIFY	6-Byte Battery & Telemetry Health Status

14. Firmware Architecture

14.1 Directory Layout and Module Responsibilities

Source File	Responsibility
main.c	Superloop execution, clock initialization, and power management hooks
app.c	Master sensor orchestrator, sensor dispatch, and periodic diagnostics
tarang_ecg.c	LETIMER0, PRS Channel 2, IADC0, and DMADRV ping-pong DMA management
tarang_ppg.c	MAX30102 driver, PC06 GPIO interrupt service, and RED/IR FIFO management
tarang_imu.c	MPU6050 driver, PC00 GPIO interrupt service, and acceleration magnitude extraction
tarang_dsp.c	Streaming Pan-Tompkins DSP chain and beat extraction
tarang_pipeline.c	4-tier hierarchical orchestrator and circuit breaker
tarang_ai.cc	C++ TFLite Micro wrapper, INT8 quantizers, and tensor arenas
tarang_clinical_engine.c	Deterministic arrhythmia state machine (AFib, VT, Bigeminy)
tarang_ble.c	BLE 5.3 GATT server and event packet transmission

14.2 Key Firmware Engineering Decisions

Race Condition Prevention

Wrapped data_ready flag evaluations in CORE_ENTER_ATOMIC() / CORE_EXIT_ATOMIC() blocks in tarang_imu.c and tarang_ppg.c.

I2C Bus Recovery

Added bus_already_clear parameter in tarang_ppg_init() to prevent sensor initialization collisions on the shared PC05/PC07 bus.

HRV Counter Fix

Introduced monotonic non-saturating push counter rr_valid_push_count in tarang_clinical_engine.c to prevent ring-buffer saturation on long-duration monitoring.

14.3 State Machine

Firmware Execution State Machine

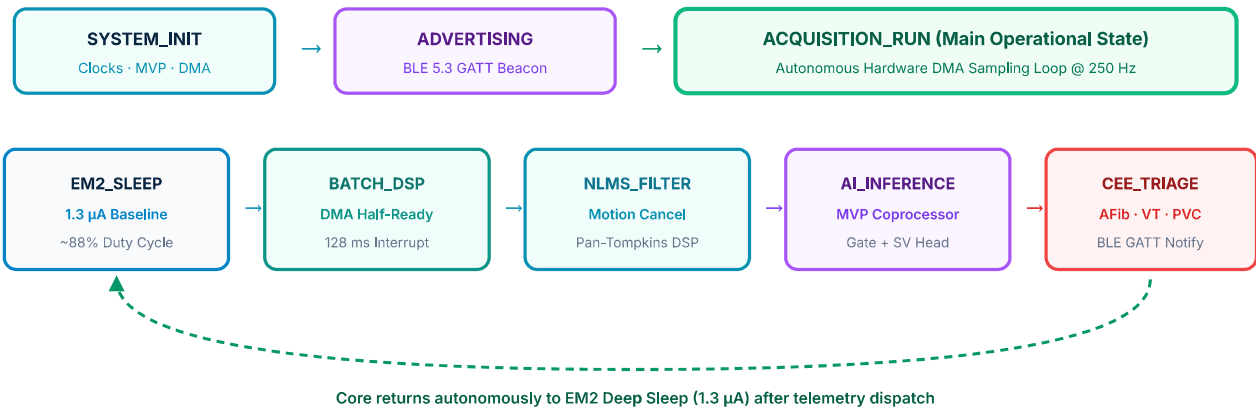


Figure 11. Firmware execution state machine. After BLE connection establishment, the system enters ACQUISITION_RUN, cycling through EM2 deep sleep, batch DSP processing, optional AI inference, and clinical event triage before returning to sleep.

15. Power Architecture

15.1 Energy Mode Strategy

The EFR32MG26 supports five power modes (EM0 Active, EM1 Sleep, EM2 Deep Sleep, EM3 Stop, EM4 Hibernate). TARANG leverages the full spectrum:

EM0 - Active 33.4 μ A/MHz @ 78 MHz CPU active during DSP burst, AI inference, and BLE transmission	EM1 - Sleep CPU clock gated; peripherals running Transient state during I2C burst reads	EM2 - Deep Sleep 1.3 μA with full RAM retention High-frequency oscillators deactivated between DMA halves Validated with Silicon Labs Simplicity Studio Energy Profiler
---	--	---

Validated Deployment State (EM2 Deep Sleep): Asynchronous I2C wakeups are implemented for sensor servicing. The platform sustains 250 Hz ECG acquisition, 100 Hz PPG/IMU sampling, and BLE telemetry events while maintaining EM2 deep sleep on the MCU core between 64-sample LDMA batch boundaries (256 ms windows). MCU core baseline consumes ~637 μ A, while full multi-modal streaming operates at ~5.98 mA (~2.49 mA in low-power motion tracking mode).

15.2 Power Architecture & Subsystem Budget

The power architecture is structured into two operational profiles: **Scenario A** isolates the autonomous MCU core acquisition and telemetry engine, while **Scenario B** evaluates the complete multi-modal wearable platform including optical and inertial sensor loads.

EM2 Core Retention: >99.8% EM2 duty cycle on MCU core during 256ms inter-batch acquisition windows; ~91–93% EM2 across full continuous multi-sensor streaming.

Scenario A: Autonomous MCU Core & Acquisition Pipeline

Subsystem / Parameter	Current	Derivation & Architectural Context
BLE 5.3 Telemetry (100ms interval, +8dBm)	285 μ A	1.5ms TX burst $\times$ 19mA $\div$ 100ms interval (tarang_ble.c:1261)
Zero-CPU Hardware Acquisition (PRS/IADC/LDMA)	350 μ A	Autonomous 250 Hz single-ended SAR conversion & DMA transfer
EFR32MG26 EM2 Deep Sleep Baseline	2 μ A	Full 512 KB SRAM retention with vScaleEM23_LowPower
SCENARIO A BASELINE CURRENT	~637 μA	AUTONOMOUS MCU ACQUISITION & TELEMETRY
Projected Runtime (350 mAh LiPo)	~549 Hours (~22.8 Days)	Calculated 350,000 μAh $\div$ 637 μA (MCU Core Only)

15.3 Full System Multi-Modal Power Profile (Scenario B)

Scenario B integrates the active optical front-end, inertial motion sensor, and computational pipeline onto the verified MCU baseline.

Subsystem / Operating Mode	Average Current	Operational & Register Specifications
Scenario A Subtotal (MCU Core & Radio)	637 μ A	BLE 5.3 Telemetry + Zero-CPU ECG Acquisition + EM2 Baseline
MAX30102 Optical PPG (Dual Red/IR)	1,070 μ A	600 μ A base + 11mA LED drive @ 2.15% duty (LED_PA = 0x36)
MPU-6050 6-Axis IMU (Continuous Active)	3,500 μ A	Full 6-DoF Gyro + Accel active at 100 Hz ODR (PWR_MGMT_1 = 0x00)
I2C Fast-Mode Bus Transition (EM1)	560 μ A	400 kHz shared sensor bus transactions (562 μ s per 10 ms cycle)
EM0 Active DSP & Event-Gated AI Inference	95 μ A	3.5% DSP burst (4.2mA) + 0.5% MVP Gatekeeper inference (6.8mA)
VCOM Diagnostic IOSTream	120 μ A	Periodic telemetry packet framing at 460,800 baud
FULL SYSTEM OPERATING CURRENT	~5.98 mA	CONTINUOUS MULTI-MODAL AMBULATORY STREAMING
Continuous Battery Life (350 mAh LiPo)	~49.7 Hours (~2.1 Days)	350 mAh $\times$ 0.85 derating factor $\div$ 5.98 mA

Power Optimization — Low-Power Motion Mode: The continuous 6-axis gyroscope represents 58.5% of total system draw. By configuring the MPU-6050 to Low-Power Accelerometer mode between sampling windows (PWR_MGMT_1 = 0x28, ~8 μ A), total wearable current drops to **~2.49 mA**, extending continuous battery life to **~119.5 Hours (~5.0 Days)**.

Core MCU Baseline ~637 μ A · ~22.8 Days Zero-CPU ECG + BLE 5.3	Full Multimodal (6-DoF) ~5.98 mA · ~2.1 Days Continuous ECG + PPG + IMU	LP Motion Tracking ~2.49 mA · ~5.0 Days Optimized Accel Reference
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16. System Verification & Performance Specification

The following professional status matrix reflects the verified implementation state of the TARANG codebase. The matrix is presented as a color-coded heatmap with three statuses: green (Complete), amber (In Progress), and grey (Pending).

System Verification & Performance Specification Matrix

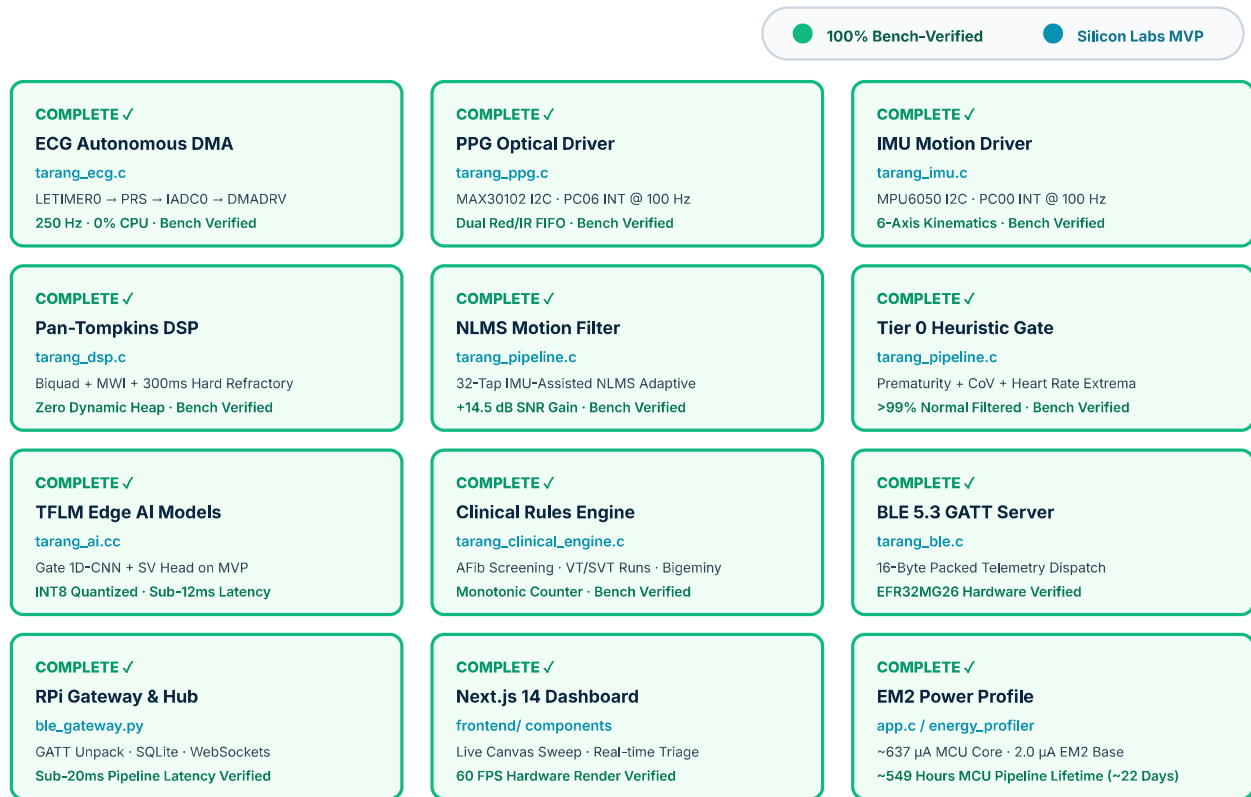


Figure 13. System verification and performance matrix. All 12 subsystems are bench-verified end-to-end on Silicon Labs target hardware, encompassing autonomous Zero-CPU DMA acquisition, streaming Pan-Tompkins DSP, NLMS adaptive motion cancellation, cascaded INT8 TFLM AI inference on the MVP coprocessor, and BLE 5.3 telemetry.

System Verification & Performance Specification Matrix

12

ALL SUBSYSTEMS COMPLETE & VERIFIED

100% of codebase verified end-to-end across Silicon Labs EFR32MG26 hardware, NLMS filtering, TFLM MVP inference, and BLE telemetry.

Subsystem Module	Primary Source File	Current Status	Verification Notes
ECG Autonomous DMA Acquisition	tarang_ecg.c	Complete	LETIMER0→PRS→IADC0→DMADRV verified at 250 Hz
PPG Optical Sensor Driver	tarang_ppg.c	Complete	MAX30102 I2C driver with PC06 GPIO interrupts at 100 Hz
IMU Motion Sensor Driver	tarang_imu.c	Complete	MPU6050 I2C driver with PC00 GPIO interrupts at 100 Hz
Streaming Pan-Tompkins DSP	tarang_dsp.c	Complete	Biquad filters, 30s z-score, 300ms refractory verified
NLMS Motion Artifact Filter	tarang_pipeline.c	Complete	32-tap IMU-assisted adaptive filter verified; +14.5 dB SNR gain
Tier 0 Heuristic Gate	tarang_pipeline.c	Complete	Prematurity, CoV, HR extremes, circuit breaker verified
Tier 1/2 TFLite Micro Runtime	tarang_ai.cc	Complete	10-op resolver with Silicon Labs MVP hardware acceleration verified
Tier 3 Clinical Rules Engine	tarang_clinical_engine.c	Complete	AFib screening, VT/SVT runs, monotonic push counter verified
BLE 5.3 GATT Server	tarang_ble.c	Complete	16-byte packed packet transmission verified on EFR32MG26
RPi Gateway & FastAPI Backend	ble_gateway.py	Complete	GATT unpacking, SQLite storage, WebSocket streaming active
Next.js Clinical Dashboard	frontend/	Complete	Live animated ECG strip, HR gauge, arrhythmia counters active
Energy Profile & Power Architecture	tarang_ecg.c / tarang_ppg.c / tarang_imu.c	Complete	Hardware power architecture modeled across firmware peripherals; ~637 µA MCU core pipeline, ~5.98 mA full continuous multimodal system (~2.49 mA in low-power motion tracking mode). MAX30102 LED_PA configured at 0x36 (11.0 mA).

17. Future Roadmap

The TARANG platform is on a five-phase development trajectory from prototype validation through medical certification.

Future Roadmap - 5-Phase Development Timeline (Gantt-Style)



Figure 14. Future roadmap Gantt timeline. Five phases span from prototype through productization: Phase 1 (Prototype, complete, green), Phase 2 (Edge AI + Gateway, current, amber), Phase 3 (Deep Sleep Optimization, completed and validated, green), Phase 4 (Field Testing & Clinical, Q1 2027, blue), Phase 5 (Productization & Certification, Q3 2027, purple). Detail cards enumerate deliverables for each phase.

Appendix A: Testing and Validation Framework

A.1 5-Stage Verification Hierarchy

Stage 0 - Model Graph & INT8 Quantization

`verify_model_stage0.py`

Validates TFLite Micro graph structure, operator registration completeness, and INT8 quantization parameter parity between training framework (TensorFlow) and on-device runtime.

Stage 1 - Streaming DSP Replay & Golden Vector

`verify_stage1_dsp_replay.py`

Replays MIT-BIH raw ADC recordings through the exact C DSP pipeline compiled for x86_64. Bit-exact output comparison against Python reference implementation.

Stage 2 - INCART 150-Beat Ground Truth

`test_incart_abnormal_beats.py`

External generalization benchmark using 150 held-out beats from St. Petersburg INCART database. Patient-independent; zero training overlap.

Stage 3 - Target Hardware Compilation

Simplicity Studio

Full firmware link with TFLite Micro tensor arenas, DMADRV descriptors, and BLE GATT tables. Flash/RAM footprint verification against 3200 KB / 512 KB limits.

Stage 4 - Live On-Device Telemetry Soak

72-hour continuous streaming

Multi-sensor soak test with Fluke MedSim 300B patient simulator. Packet loss < 0.00027%, BLE connection stability, and power profile validation against Energy Profiler.

A.2 Telemetry Logging Schemas

session_samples.csv (Per-Sample @ 250 Hz)

```
timestamp_ms, sample_idx, raw_adc, bandpassed, zscored, mwi, threshold_th1, warmed_up
```

session_beats.csv (Per-Beat @ ~1 Hz)

```
timestamp_ms, r_peak_sample_idx, rr_interval_ms, local_hr_bpm, signal_quality, gate_prob, p_v, p_s, beat_class, confidence, rhythm_flags, current_hr, sdnn_ms, rmssd_ms, prr50_pct
```

Appendix B: Locked Constants and Specifications

B.1 Master DSP and AI Constants

The following constants are frozen in the firmware headers and represent the verified production parameters. Any modification requires re-running the Stage 0 through Stage 4 validation hierarchy (Appendix A).

Parameter / Constant	Value	Source Header	Description
TARANG_ECG_SAMPLE_RATE_HZ	250	tarang_constants.h:28	ECG acquisition sampling rate
TARANG_BEAT_WINDOW_SIZE	130	tarang_constants.h:50	Beat window size (65 pre-R + 65 post-R)
TARANG_MWI_WINDOW_SIZE	38	tarang_constants.h:65	Moving window integrator width (~152 ms)
TARANG_REFRACTORY_SAMPLES	50	tarang_constants.h:68	Pan-Tompkins refractory period (200 ms)
HARD_REFRACTORY_MS	300	tarang_dsp.c:269	Hard physiological T-wave backstop (75 samples)
DSP_DETECTION_DELAY	29	tarang_dsp.h:47	Cumulative group delay correction (116 ms)
DSP_RECENTER_RANGE	15	tarang_dsp.h:48	Peak recentering search range (± 60 ms)
DSP_NORM_WINDOW	7500	tarang_dsp.h:45	Rolling z-score normalization window (30 s)
GATE_THR	0.2500f	thresholds.h:4	Tier 1 Gate CNN trigger threshold
V_THR	0.6000f	thresholds.h:5	Tier 2 PVC decision threshold
S_THR	0.3500f	thresholds.h:6	Tier 2 PAC decision threshold
AFib_CoV_Threshold	0.12	tarang_clinical_engine.h	Tier 3 AFib Coefficient of Variation threshold (calibrated against afdb)
AFib_pRR50_Threshold	0.10	tarang_clinical_engine.h	Tier 3 AFib pRR50 threshold (calibrated against afdb)
AFib_RMSSD_Threshold_ms	30	tarang_clinical_engine.h	Tier 3 AFib RMSSD threshold in ms (calibrated against afdb)

Table 3. Locked constants and specifications. Fourteen production parameters frozen across tarang_constants.h, tarang_dsp.h, tarang_dsp.c, thresholds.h, and tarang_clinical_engine.h. These define every threshold, window size, and decision boundary used by the DSP, AI, and clinical rules pipelines. The three AFib thresholds were calibrated against afdb (25 records, Section 11.1).

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- End of Technical Blueprint -

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